

ANDROID APP DEVELOPMENT

By

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STUDENT DECLARATION

I hereby certify that this dissertation constitutes my own product, where the language of others is set forth, quotation marks so indicate, and that appropriate credit is given where I have used the language, ideas, expressions or writings of another.

I confirm that I have not copied material from another source nor committed plagiarism nor commissioned all or part of the work (including unacceptable proof-reading) nor fabricated, falsified or embellished data when completing the attached piece of work.

I declare that the dissertation describes original work that has not previously been presented for the award of any other degree of any institution.

Signed,

Sanket Churi

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Android App Development

Reservation Table Handler

Abstract

The following is a dissertation on the creation of a table management application for Android that is designed to overcome shortcomings in restaurants especially during busy periods with a lot of reservations. This has been greatly attributed by observing that control of the tables during peak periods was difficult while working for the restaurant business. Such complications manifested themselves in delays, double bookings, and cross communication among the workers which impacted the customers' experience negatively. This project was undertaken to find a solution that will help minimize the challenges associated with table assignment, poor communication among the staff and poor customer care during busy hours.

This Android application was developed using the Android Studio tool; and the Agile development methodology was employed in the manufacturing process of the android application so that feedback from the restaurant staff would be incorporated at every stage in the production of the app. This approach made it possible to develop an application that meets the dynamism characteristic of the restaurant business environment. It has some neat features such as monitor table in real time and book a table instantly, send notification to the concerned person so that there wouldn't be a problem in reservation of two different people for the same table. Java/Kotlin was used in the development of the backend of the application while XML was used in the development of the interface that would not be very hard to master for the restaurant workers.

The testing of the application was carried out in a restaurant environment where scenarios were created with close resemblance to real working conditions. Encouragement of the staff members' feedback was done to enhance the application further in terms of its function. The outcome pointed to a more optimized table allocation process and brought down communications errors and overlaps, especially during the busy hours. Employees also stated that using the app, booking management became easier and team members could also align better and deliver a better service to the customers.

It is finally established in this study that the application of a proper table management system could reduce work productivity loss during rush hours in restaurants. Further developments could be made by embedding this application into currently used restaurant management software and including a complementary analytical layer; one that will predict the times of high traffic and optimize table IDs turnover rates. This project provides one practical idea which restaurants face in their everyday business and demonstrates how technology would be a solution for staff, increasing efficiency, avoiding mistakes, and increasing satisfaction levels. By adopting such technologies, restaurants can be able to manage the rush hour, customer service delivery and enhance the experience of their employees and customers.

Statement of Ethical Compliance

Data Category: - A

Participant Category: - 2

I confirm that the app which I am developing named Reservation Table Management App will be aware of all the ethical guidelines set forth by the University of Liverpool. The app will be designed to handle table reservations at restaurants to ensure that there is no overbooking done. It will manage non-sensitive data related to the reservation slots and customers details without collecting any personal information except basic reservation details. If there are any further changes in the application, it will be reviewed with the supervisor's guidance to maintain ethical standards.

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Chapter 1: Introduction

1.1 Scope of the Study

An Android application which would help in better management of table in a restaurant set up is the main concept of this work. The app will seek to solve issues that are likely to be rampant during the rush hours in a restaurant for instance, wrong seating layout, delayed service delivery, and poor communication between the employees. With major features such as the real timetable status, reservation and feedback options, the app aims at minimizing the complexity of the dining process for the customers as well as the employees of the restaurant.

1.2 Problem Statement

Customers generate a lot of inconvenience and loopholes to restaurants especially on table management during the busy periods of the day. Key issues identified include:

- Table allocation is ineffective, leading to long queues or a long waiting time.
- Staff do not work in unison, hence the possibility of having to reserve two tables or having tables with excess capacity.
- Some challenges are proving a challenge in managing both the reservations and the walk-ins.
- Lack of communication between front of house employees- host, waiter and back of house employees- chefs and cooks.

All these issues are worse during rush hour, resulting in a confusing eating pattern, staff pressure, and poor customers' feedback. Lack of a proper system of managing the table stock as well as the number of reservations comes as one of the major operational losses (Vidotto et al., 2007).

1.3 Approach

The approach of this study involves the development of a mobile application using Android platform technologies. The app will include the following key features:

1. **Real-Time Table Management:** A dynamic interface displaying table statuses (occupied, available, reserved) to help hosts and waitstaff make quick decisions.
2. **Reservation System:** Integration of a reservation management module that allows customers to book tables in advance, with notifications sent to staff.
3. **Waitlist Management:** A feature for adding walk-in customers to a waitlist, with estimated wait times calculated based on current table occupancy.
4. **Staff Coordination:** Internal messaging capabilities to facilitate better communication between staff members, reducing errors and improving service speed.

The development process will utilize agile methodologies, including iterative design, user feedback loops, and testing phases to refine the app's functionality and ensure it meets the needs of the restaurant environment.

1.4 Expected Outcomes

The expected outcomes of this study include:

1. Improved Table Turnover: Enhanced efficiency in managing tables will lead to faster table turnovers, increasing the restaurant's capacity to serve more customers during peak hours.
2. Reduced Waiting Times: Real-time updates and a clear reservation system will help manage customer expectations and reduce wait times.
3. Enhanced Customer Satisfaction: A more organized and transparent seating process will improve the overall customer dining experience, potentially increasing repeat visits and positive reviews.
4. Better Staff Coordination: The app will facilitate better communication among staff members, reducing errors and stress during high-demand periods.(Vidotto et al., 2007)

Chapter 2: Introduction and Background

2.1 Introduction

This chapter reviews the existing literature and background relevant to restaurant management systems, table management during peak shifts, and the development of mobile applications in the hospitality industry. Given the absence of existing solutions tailored to the specific challenges identified in restaurant table management, this chapter primarily draws on practical sources such as websites and online tutorials to inform the development of the app.

2.2 Overview of Restaurant Management Challenges

Restaurant management, particularly during high-traffic periods, presents several challenges, including table allocation inefficiencies, communication breakdowns among staff, and extended customer wait times. These issues are well-documented across industry websites, forums, and restaurant management blogs, highlighting the need for effective solutions to streamline operations and enhance customer service during peak hours (DiPietro, 2017).

2.3 Existing Solutions and Their Limitations

Existing technologies such as traditional POS systems and reservation management apps provide some support for restaurant operations. However, a review of practical sources, including software review websites and user feedback platforms, reveals significant gaps. These systems often lack real-time data integration, have limited customization options, and present usability challenges that hinder staff efficiency during busy shifts. Despite their presence, none fully address the nuanced needs of table management observed in the restaurant environment (Vidotto et al., 2007).

2.4 Learning from Online Resources

During the development process various online resources such as YouTube.com (<https://www.youtube.com/watch?v=04bzIg3Xuv&t=1s>), helped overcome challenges related to Kotlin and XML layout design. Specifically, a tutorial by **Google Developers** [https://developer.android.com/courses/android-basics-compose/unit-1?_gl=1*_hph5t6*_up*MQ..*_ga*ODYwMTkwNzAuMTcyNjc2NjY2Ng..*_ga_6HH9YJMN9M*MTcyNjc2NjY2Ni4xLjAuMTcyNjc2NjY2Ni4wLjAuNzcwODAzMTE2_] on integrating Android architecture components was essential for managing data persistence.

2.5 Practical Implementation Challenges

Throughout the app development process, several challenges were encountered, including coding hurdles, feature integration, and user interface design tailored to

restaurant staff needs. Online resources served as a vital tool in navigating these challenges, providing direct examples and problem-solving techniques that were adapted to suit the unique context of restaurant table management.

2.6 Justification for App's Development

The primary motivation behind developing this app was the observed gap between existing tools and the real-world needs of restaurant staff during heavy shifts. Current solutions did not adequately address the dynamic requirements of table management, leading to inefficiencies and operational delays.(Hwang, 2008) The app was therefore designed with features specifically tailored to mitigate these issues, such as real-time updates, easy table allocation, and an intuitive interface that supports rapid decision-making.(Du et al., 2021)

2.7 Gaps in Existing Online Resources

While online tutorials and websites were invaluable, there were notable limitations in the available content, such as missing advanced guidance on implementing hospitality-specific features or outdated practices that required adaptation. These gaps necessitated a degree of innovation and problem-solving beyond what was directly available, reinforcing the uniqueness of the developed app.

2.8 Summary

This chapter has reviewed the background and practical literature relevant to restaurant management during peak periods, emphasizing the limitations of existing systems and the practical resources used in app development. The findings underscore the necessity of a customized approach to table management, as reflected in the app developed through direct observation, practical learning, and adaptation of online resources.

Chapter 3: Design

3.1 Original Design

The app's design was initially driven by the practical needs observed in a restaurant setting, focusing on improving table and booking management to enhance operational efficiency during busy periods. The original concept centered on providing a seamless interface for staff to manage reservations, table assignments, and real-time updates on table availability.

3.1.1 Key Features

1. **Booking Management:** A comprehensive booking management system was implemented, allowing users to create, update, and delete bookings easily. The system supports input fields for customer details, number of guests, table assignment, date, time, and additional notes.
2. **Dynamic Table Management:** The app allows the management of tables within the restaurant. Staff can add or remove tables, view their status, and update seating arrangements dynamically.
3. **User-Friendly Interface:** The design included a simple, intuitive interface that minimizes training time for staff. Key functions are accessible via a bottom navigation bar, with large, clearly labeled buttons for booking management, dashboard, and settings.

3.2 Design Changes

As the app was developed and tested in real-world scenarios, several changes were made to the initial design to better suit the needs of the restaurant environment.

3.2.1 Implemented Changes

1. **Enhanced Booking System:** Initially, the booking system allowed only basic bookings. Changes included adding support for displaying bookings in a visually organized format, grouping bookings by date, and formatting times and dates to be more readable for users.
2. **User Feedback Integration:** Based on feedback from restaurant staff, additional functionalities such as the ability to expand booking details, manage notes more effectively, and customize table assignments were added. This helped reduce errors during table assignments and provided a clearer view of customer requirements.
3. **Date and Time Selection Improvements:** The date and time selection components were enhanced to provide a more user-friendly interaction, including minimum date settings to avoid past bookings and time formatting changes that reflect the restaurant's operational hours.
4. **Backup and Data Security:** Recognizing the importance of data security and integrity, a background backup feature was added to periodically save booking and table data, ensuring that no critical information is lost in case of a device failure or data corruption.

3.3 Rationale for Changes

1. Improved Usability and Staff Efficiency: Enhancing the booking display and formatting made it easier for staff to quickly understand customer needs and manage bookings without confusion, directly addressing feedback received during testing.
2. Flexibility in Table Management: Allowing dynamic updates and visual representations of table layouts was essential for accommodating the constantly changing set-up of the restaurant, especially during events or peak hours.
3. Enhanced Customer Experience: Adding features to manage bookings with more granularity, such as expanding notes and easily updating booking details, improved the overall customer experience by reducing wait times and ensuring accurate information flow among staff (Hwang, 2008).
4. Data Integrity and Security: Implementing data backups and role-based access controls addressed the need for secure handling of sensitive booking data, ensuring that only authorized personnel could make critical changes.
5. User-Centric Enhancements: By incorporating direct feedback from users, the design evolved to meet the practical demands of the environment, making the app a reliable tool for managing restaurant operations under varying conditions.

Chapter 4: Implementation and Design

4.1 UI/UX Design

4.1.1 Implementation

The implementation phase of the Table Reservation Handler app involved a series of steps, from setting up the development environment to coding the core features such as table availability, booking management, and real-time notifications. The app was designed using **Android Studio**, with **Kotlin** as the primary programming language due to its interoperability with Java and its ability to reduce boilerplate code, making it a more efficient choice for Android development.

4.1.2 Development Platform

The app was developed using **Android Studio**, the official IDE for Android app development. Android Studio offers several tools, such as a layout editor, an emulator, and built-in debugging features, which facilitated the development process. It also supports **Kotlin**, which was used for the implementation of both front-end and back-end features. Android Studio's integrated **Gradle** build system helped manage dependencies, compile the app, and ensure smooth deployment across various devices.

4.1.3 Programming Language

The decision to use **Kotlin** instead of Java was based on Kotlin's modern features, such as null safety and extension functions, which reduce the likelihood of runtime errors. Kotlin's concise syntax significantly reduces the amount of code required, making it easier to maintain and debug, while still being fully interoperable with Java.

4.2 Database Design and Implementation

To support the functionality of the Reservation Table Management application, a relational database was implemented using **SQLite**, a lightweight yet powerful database engine suited for mobile platforms like Android. The database stores and manages critical information related to customer bookings and restaurant tables. This section outlines the structure and key operations of the database.

4.2.1 Database Schema

The database consists of two primary tables:

1. **Bookings Table:** This table stores all the reservation data, including customer information, the number of guests, booking time, table assignment, and optional notes.
2. **Tables Table:** This table manages information about the tables available in the restaurant, including each table's number and capacity.

The database schema was defined as follows:

- **Bookings Table:** The bookings table consists of the following fields:
 - id: A unique identifier for each booking.
 - customer_name: The name of the customer making the booking.
 - phone_number: The customer's contact number.
 - guest_count: The number of guests in the reservation.
 - table_number: The table assigned to the reservation.
 - date: The date of the booking.
 - time: The time of the booking.
 - notes: Any special requests or additional information for the booking.
- **Tables Table:** The tables table contains:
 - id: A unique identifier for each table.
 - table_number: The number of the table.
 - table_capacity: The seating capacity of the table.

These tables are created during the initialization of the database and managed by the **RestaurantDbHelper** class. The schema ensures that each booking is properly linked to a specific table, with additional fields to manage guest information and notes. This relational structure allows for efficient queries and data management within the application.

4.2.2 CRUD Operations

The database operations are handled via standard **Create, Read, Update, and Delete (CRUD)** operations. These operations are essential for managing both bookings and tables dynamically in the application. For example, when a customer makes a new booking, the following steps occur:

- **Create:** A new entry is inserted into the bookings table, including details such as the customer's name, phone number, number of guests, and table number.
- **Read:** The application can query the database to retrieve a list of all bookings or available tables, ensuring real-time updates for restaurant staff.
- **Update:** If a booking needs to be modified (e.g., changing the number of guests or table assignment), the system can update the existing entry in the database.
- **Delete:** When a booking is canceled, the corresponding entry is deleted from the bookings table, freeing up the table for future reservations.

The **insertBooking** method is responsible for inserting a new booking into the database. It uses ContentValues to store the booking details and inserts them into the database. Similarly, the **getAllBookings** method retrieves all bookings from the bookings table and orders them by date and time.

```

// Bookings methods (like insertData, getAllBookings) go here
fun insertBooking(customerName: String, phoneNumber: String, guestCount: Int, tableNumber: String?, date: String, time: String) {
    val db = this.writableDatabase
    val contentValues = ContentValues().apply {
        put("customer_name", customerName)
        put("phone_number", phoneNumber)
        put("guest_count", guestCount)
        put("table_number", tableNumber)
        put("date", date)
        put("time", time)
        put("notes", notes) // Added line
    }
    db.insert( table: "Bookings", nullColumnHack: null, contentValues)
}

```

Figure 1 *insertBooking* Function

4.2.3 Managing Table Availability

One of the core features of the app is its ability to manage table availability in real-time. To achieve this, the application checks if any tables are reserved within a specific time buffer (typically one hour before and after the booking time). The **getAvailableTables** method queries the database to retrieve a list of tables that are not reserved during a given period, ensuring that staff can make quick decisions when assigning tables to new reservations.

The application employs **date and time formatting** to compare booking times and determine table availability. The following steps are taken:

1. The selected booking time is parsed and adjusted to calculate a one-hour buffer.
2. A query is executed to check for any bookings that overlap with this time buffer.
3. The system returns a list of available tables, which are displayed in the user interface for staff to choose from.

Chapter 5: Testing and Evaluation

Rigorous testing is a crucial part of the development lifecycle for any software application, especially one that will be used in a fast-paced environment like a restaurant. To ensure the reliability and usability of the Table Reservation Handler app, a combination of **Unit Testing**, **Integration Testing**, **System Testing**, and **User Acceptance Testing (UAT)** was employed throughout the development process. These tests helped identify bugs, optimize performance, and refine the user experience to meet the needs of restaurant staff.

5.1 Unit Testing

Unit Testing focuses on verifying that individual component of the app work correctly in isolation. For the Table Reservation Handler app, each key function—such as booking creation, table availability updates, and waitlist management—was tested to ensure it performed as expected under various conditions.

The reservation system is one of the app's core functionalities, and unit tests were designed to ensure that:

1. **Reservation Creation:** When a user creates a reservation, the app properly checks table availability based on the selected time and updates the reservation in the SQLite database. This prevents double bookings by locking the table during the specified time slot.
2. **Reservation Modification and Cancellation:** Tests were conducted to verify that modifying an existing reservation correctly updates the table's status in the database. Additionally, canceling a reservation should free up the table immediately, making it available for future bookings.
3. **Reservation Time Constraints:** Each reservation has an estimated dining time associated with it. Tests ensured that the app calculates and respects this time constraint to prevent overlapping bookings. The system was checked to ensure that a table marked as occupied cannot be reserved again until the current reservation is complete.

5.2 Integration Testing

Integration Testing ensures that the individual components of the app work together cohesively. Since the Table Reservation Handler app consists of several modules, such as the front-end user interface, the SQLite database, and the notification system, integration tests were essential to validate the interactions between these parts.

5.2.1 Front-End and Back-End Communication

- **Booking Interface and Database:** Integration tests were carried out to verify that actions taken in the UI—such as creating, modifying, or canceling reservations—correctly updated the database. These tests involved:

- Checking that new reservations were properly saved to the SQLite database.
- Ensuring that any changes made to existing reservations were accurately reflected in the database and the front-end UI.
- Verifying that when reservations were canceled, the table's availability was immediately updated in the UI and stored in the database.
- **Table Status Synchronization:** Since the app displays real-time table availability, the front-end interface was tested to ensure it remained synchronized with the back-end database. This included checking that tables marked as reserved or occupied in the database were reflected as such in the user interface.

5.3 System Testing

System testing focuses on validating the entire app in a real-world scenario. The Table Reservation Handler app was tested on various Android devices with different screen sizes, resolutions, and operating systems to ensure that it worked as intended in all environments. This type of testing also helped ensure that the app could handle different load conditions, such as high reservation traffic during peak hours.

5.3.1 Cross-Device Compatibility

The app was tested on various Android devices to ensure compatibility, including:

1. **Different Screen Sizes and Resolutions:** The app was tested on smartphones and tablets to verify that the UI adapted properly to different screen sizes and resolutions. This included ensuring that the dashboard, reservation forms, and table setup screens were fully responsive.
2. **Android Versions:** The app was tested across different Android versions, from older versions like Android 8.0 (Oreo) to newer versions such as Android 12, to ensure backward compatibility and smooth operation across devices.

5.3.2 Performance Testing

The app's performance was assessed under high-stress conditions to ensure that it could handle many bookings and updates without slowing down or crashing. This included:

- **Load Testing:** Simulating heavy usage, such as multiple reservations being created simultaneously, was performed to ensure the system could maintain its performance without becoming sluggish.
- **Memory and Resource Usage:** The app's memory consumption was monitored to ensure it remained within acceptable limits, even when handling multiple tasks (e.g., updating reservations, managing waitlists, and sending notifications).

5.3.3 Real-World Scenarios

Real-world scenarios were simulated to test how the app responded to practical use cases:

- **Simultaneous Booking and Walk-ins:** The system was tested by creating reservations for multiple customers while also adding new walk-ins to the waitlist. This scenario helped ensure that the app could manage bookings and notify staff of table availability in real-time, even under heavy load.
- **Internet Connectivity Issues:** Since the app relies on real-time updates, it was tested to see how it handled scenarios with intermittent internet connectivity. The app was designed to function offline for local table management and synchronize with the cloud once the internet connection was restored.

5.3.4 Conclusion

Through comprehensive testing—including unit, integration, and system testing—the Table Reservation Handler app was optimized for both performance and usability. By incorporating feedback from real users during the UAT phase, the app was refined to meet the practical needs of restaurant staff, ensuring that it performed reliably under real-world conditions. These testing phases ensured that the app was robust, user-friendly, and capable of handling the operational demands of busy restaurant environments.

Chapter 6: Ethical Considerations

The ethical considerations in the development of the Table Reservation Handler app were carefully evaluated to ensure the protection of data, the rights of the participants involved, and adherence to ethical guidelines. This section outlines the ethical approach taken during the app's development, including the involvement of human participants, data privacy measures, and the overall ethical management of the project.

6.1 Project Ethics and Human Participants

Human participants were directly involved in the testing and feedback phases of the app's development. The primary aim of involving restaurant staff and managers was to gain insights into their operational challenges, collect feedback on the app's usability, and identify areas for improvement. Ethical practices were strictly followed to ensure that participants' rights and data were respected throughout the process.

6.2 Usability Testing

Usability testing involved restaurant staff using the app in a real-world environment to evaluate its effectiveness, ease of use, and overall functionality. This process was critical in refining the app to meet the specific needs of its target users. The participants tested the app during their normal shifts, and their feedback was gathered through:

- **Direct Observation:** Observed how staff interacted with the app, noting any difficulties or inefficiencies they encountered.
- **Feedback Sessions:** After using the app, participants provided structured feedback on the app's performance, including its interface design, responsiveness, and usability during peak hours.

The feedback from usability testing helped refine key aspects of the app, including simplifying the booking process, improving navigation, and enhancing the visual layout of the table management dashboard.

6.3 Ethicalness of the Project

The project adhered to ethical guidelines throughout the development process, with specific emphasis on obtaining informed consent, maintaining transparency, and ensuring that the rights of participants were protected.

6.3.1 Informed Consent

Informed consent was a foundational ethical principle in this project. Before participating in interviews, usability tests, or feedback sessions, all respondents were:

- **Informed of the Objectives:** Each participant was clearly informed of the purpose of the project, the nature of the app being developed, and how their feedback would be used to improve the app. The objectives of the research were communicated in both written and verbal form.
- **Voluntary Participation:** Participation in the project was entirely voluntary. Respondents were made aware that they could withdraw at any point without facing any negative consequences. This ensured that no one was coerced or pressured into participating in the research process.

All participants signed a **Participant Consent Form**, acknowledging that they understood the objectives of the research and agreed to participate willingly. Copies of the signed forms were securely stored and submitted to the University for record-keeping purposes

6.4 Summary of Ethical Considerations

The development of the Table Reservation Handler app was carried out in accordance with the highest ethical standards. By obtaining informed consent, involving human participants in a transparent and voluntary manner, and ensuring the confidentiality of all data collected, the project maintained ethical integrity throughout. Regular supervisory oversight ensured that all research activities were conducted responsibly, with full consideration given to the rights and privacy of the participants.

Chapter 7: Learning Points

Throughout the development of the Table Reservation Handler app, several key learning points emerged, both technical and non-technical. These lessons were integral not only to the success of the project but also to the personal development of the developer. This section highlights the most significant areas of learning, from mastering new technologies to navigating the challenges of app development in a real-world context.

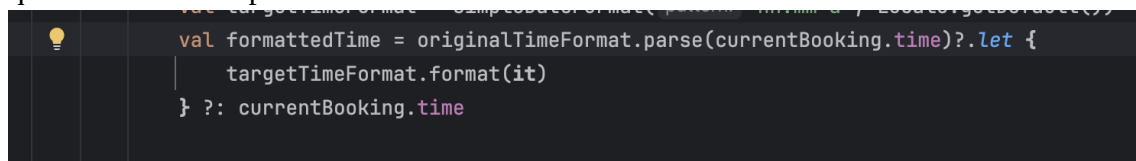
7.1 Technical Skills and Development

The process of building the Table Reservation Handler app required acquiring and refining various technical skills. These skills were developed through practical experience, problem-solving, and self-directed learning from online resources, tutorials, and documentation.

7.1.1 Kotlin for Android Development

Prior to this project, knowledge of Kotlin was limited, but through the development process, a deep understanding of Kotlin was gained. Kotlin was chosen for its modern syntax, null safety, and seamless interoperability with Java, making it an ideal choice for Android development. Several key features of Kotlin were essential to the app's development:

- **Conciseness and Readability:** Compared to Java, Kotlin's concise syntax helped reduce the amount of boilerplate code, improving code readability and maintainability.
- **Null Safety:** Kotlin's built-in null safety feature was instrumental in avoiding common bugs related to null references, particularly when handling database queries and user inputs.



```
val formattedTime = originalTimeFormat.parse(currentBooking.time)?.let {  
    targetTimeFormat.format(it)  
} ?: currentBooking.time
```

Figure 2 Null Handling

7.1.2 Android Studio and Android Architecture Components

Another significant learning point was mastering **Android Studio** and various Android architecture components such as:

- **ViewModel:** Managing UI-related data in a lifecycle-conscious way to ensure data persists during configuration changes, such as screen rotations.

- **LiveData:** Observing data changes and updating the UI in real-time without manual intervention.
- **Room Database:** Using Room as a database management layer on top of SQLite simplified database operations, enabling structured queries while maintaining an object-oriented approach.

Learning to navigate Android Studio's extensive features—including the built-in emulator, debugging tools, and layout editor—was crucial in efficiently developing the app. Understanding how to use Gradle to manage dependencies and optimize the build process was also a key technical milestone.

7.1.3 Database Management with SQLite

Working with **SQLite** for local data storage was a critical learning point, as it was necessary to store and retrieve table and reservation data efficiently. The process involved designing database schemas, writing SQL queries, and optimizing data retrieval to ensure real-time table availability updates without performance bottlenecks.

Key takeaways from this experience include:

- **Database Schema Design:** Structuring tables in a way that minimized redundancy while ensuring scalability as more reservations were added.
- **Query Optimization:** Writing efficient SQL queries to handle large datasets without slowing down the app's performance, especially during peak hours when reservations would be at their highest.

Additionally, learning to handle database migrations in SQLite to ensure that the app could be updated without losing important user data was another technical challenge that was overcome during the development process.

7.2 Problem Solving and Debugging

App development is rarely without its challenges, and this project provided numerous opportunities to develop problem-solving skills. Debugging errors, integrating features, and optimizing performance required a methodical approach and a willingness to experiment with different solutions.

7.2.1 Debugging and Error Handling

Early in the development process, several bugs related to database operations and UI behavior emerged. For example:

- **Database Insert Failures:** Initial attempts to insert reservation data into the database occasionally failed due to issues with null values or mismatched data.

- types. Through debugging tools in Android Studio, these issues were diagnosed, and Kotlin's null safety features were leveraged to ensure robust data handling.
- **UI Rendering Issues:** When implementing real-time table availability updates, the app sometimes experienced delays in updating the UI. Through trial and error, the issue was traced to the improper use of threads in handling background tasks. Learning to use Kotlin's **coroutines** to handle asynchronous operations smoothly resolved these issues, significantly improving performance

7.3 User-Centered Design and Usability

One of the most valuable lessons learned from the development of the Table Reservation Handler app was the importance of **user-centered design**. Since the app was intended for use by restaurant staff, most of whom are non-technical, the design needed to prioritize simplicity and ease of use. The feedback gathered during **User Acceptance Testing (UAT)** provided crucial insights into how the app could be improved to better meet user needs.

7.3.1 Importance of Intuitive Design

The initial version of the app, while functional, was found to have certain complexities in the UI that slowed down staff during peak hours. Based on the feedback from restaurant staff:

- **Simplifying the Reservation Form:** The reservation form was simplified by automatically populating certain fields (such as the current date and time) to reduce the number of steps required to complete a booking.
- **Improving Navigation:** Some users reported difficulty navigating between different sections of the app. The solution involved adding more intuitive icons and simplifying the layout to ensure that important actions—such as checking table availability—were always just one tap away.

7.4 Time Management and Project Planning

Effective time management and project planning were also critical learning points during the development process. The initial project timeline set out specific goals and milestones, but as the project progressed, certain tasks—such as debugging and refining the UI—took longer than expected. This required the project plan to be adjusted to accommodate unforeseen challenges.

7.4.1 Importance of Flexibility in Project Timelines

One of the key lessons was the need to build flexibility into project timelines. While having a structured plan is important, it's equally important to account for potential delays. As new challenges arose—such as unexpected bugs or additional user

requirements—the timeline had to be revised to ensure that the project stayed on track without compromising quality.

7.5 Summary

The development of the Table Reservation Handler app was a valuable learning experience that extended beyond technical skills. From mastering new programming languages and tools to navigating the complexities of user-centered design and time management, the project provided a comprehensive foundation for future software development. The combination of technical challenges, real-world feedback, and project planning lessons will serve as a strong base for future endeavors in app development.

Chapter 8: Future Versions

- **Enhancements and Future Features:**
 - **Integration with POS Systems:** "Future versions of the app could include integration with existing Point-of-Sale (POS) systems to streamline operations further, enabling automatic updates to table status based on orders and payments."
 - **Advanced Analytics:** "Adding analytics capabilities could help restaurant managers predict peak times, optimize table turnover, and make data-driven decisions to improve efficiency."
 - **Multi-Platform Support:** "Expanding the app to iOS and web platforms would make it more versatile and accessible, catering to a wider audience of restaurant staff who might use different devices."
 - **Individual Login:** "Staff would have their personal login id's which can be helped to identify who has registered the table along with only manager will have the rights to edit or delete entered data"
- **Scalability Improvements:**
 - As the app scales, moving to a more robust backend such as Google Cloud Fire store or AWS could handle higher data loads and offer better performance for larger establishments.

Chapter 9: BCS project criteria

1. Practical and Analytical Skills Application:

- Design easy-to-use UI/UX for the restaurant staff.
- Efficient Algorithm implantation is done so that reservations can be handled dynamically regarding cancellation, walk-in and no-show ups.

2. Invention and/or Originality Invention:

- Real-time updates on availability of staff or analytic dashboards, automated notifications sent to the staff.
- Originality: User-friendliness on the interfaces and the interactions, e.g. drag and drop UIs for designed table arrangements or visualizations of the reservation statuses.

3. Synthesis of Information:

- Ideas, and Practices Requirements are gathered by interviewing needs of people working in a restaurant are to have a system that can be used in various restaurants with the following configuration.
- The scalability and adaptability of the app to be used in various restaurants.
- Requirements are met for the system and include that it is usable in all restaurants with only a minimum configuration.

4. Meeting a Real Need in a Wider Context Need:

- Very few Restaurants avoid over-booking which.
- Context: A solution would not only be useful for my restaurant or just one environment; it would also be relevant to many others who face similar problems on their site and thus, of wide applicability and marketability

5. Management of a Major Piece of Work:

- Documenting, in a clear fashion, my plan with milestones, deliverables, and timelines. Effectively use time to balance development, testing, and feedback phases.
- Development from the first concept to the last deploy after testing serves as proof that I can work on a big project independently.

6. Critical Self-Appraisal of the Process:

- By keeping a constant check on my progress and the effectiveness of my developed app, I shall understand where I can make improvements and document what I have learned and then update my development approach accordingly.
- Reviews post-deployment: Assess how an app has performed under real-use conditions and user feedback for future enhancements.

Chapter 10: Conclusion

The design of the app reflects an iterative approach, driven by real-world testing and feedback from its end-user's restaurant staff. The changes made throughout the development process enhanced the usability, flexibility, and overall effectiveness of the

app, aligning with the goal of streamlining tables and booking management in a busy restaurant environment. The design not only addresses the initial challenges identified but also adapts to the evolving needs of its users, making it a valuable addition to the restaurant's operational toolkit.

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Appendix

Consent Forms



Department of Computer Science
Ashton Street, Liverpool, L69 3BX

Computer Science Project Modules Participant Consent Form

Project Title: Android Application Development (Reservation Table Handler)
Student: Sanket Manoj Churi
Supervisor: Jimmieson Phil
Coordinator: Dr Stuart Thomason (s.thomason@liverpool.ac.uk)

Instructions to student: Mark the boxes that apply to your project so the participant knows what you will be asking them to do. If you later decide to ask the participant to do something that isn't marked, fill in a new form.

You are inviting the participant to take part in... (Copy/paste the ☒ symbol where necessary)

☐ Requirements Analysis ☒ Software Evaluation

You plan to use the following methods as part of this activity...

☐ Interview ☐ Discussion ☒ Observation
☐ Narration ☐ Questionnaire

You will be using the following techniques to record what the participant is saying or doing during the activity...

☐ Audio Recording ☐ Screen Recording ☐ Written Notes
☐ Apple TestFlight or Google Console (for which the participant's Apple or Google ID will be required)

Instructions to participant: Please read the Participant Information Sheet carefully. If you agree to take part in the activities indicated above, enter your initials in the box below and return this sheet to the student.

- 1) I confirm that I have read and understood the information sheet attached to this form, or it has been read to me. I have had the opportunity to consider the information and ask questions. My questions have been answered to my satisfaction.
- 2) I understand that my participation is voluntary and that I am free to stop taking part at any time without giving a reason. In addition, I understand that I do not have to answer all questions or take part in all listed activities.
- 3) I understand I can ask to see the information I have provided, and I can ask for it to be destroyed before it is aggregated and anonymised. I understand that after the information has been anonymised, I can no longer ask for it to be destroyed.
- 4) I understand that the information I provide will be stored in line with the data protection policy of the University of Liverpool until it is destroyed.
- 5) I understand that my personal contribution will be destroyed when it has been aggregated and anonymised.
- 6) I agree that this consent form will be retained by the project supervisor for a period of two months following submission of the dissertation of which it forms part, after which it will be destroyed.

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Consent Form for Participation in Video Presentation

Project Title: Table Reservation Handler (Android Application) - Dissertation Project

Researcher: Sanket Manoj Churi

Student ID: 201700900

Dear Participant,

Thank you for taking part in my project video presentation. This project's purpose is to demonstrate the functionality of the Table Reservation Handler app, which aims to improve the table booking process in restaurants.

Consent: By signing below, you acknowledge and agree to the following:

1. You voluntarily agree to participate in this project and appear in the video presentation.
2. You understand that your involvement is for academic purposes, specifically my dissertation project.
3. You consent to the use of your feedback on the app and your appearance in the video.
4. Your feedback will be used to improve the app and will be reported anonymously in my dissertation.

Participant Name: Elisha Wilkerson-Forster

Signature: Elisha

Date: 30th August 2024

Thank you for your valuable contribution!

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Participant Name: Fabiana Hespanhol

Signature: Fabiana

Date: 30th August 2024

Thank you for your valuable contribution!

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Participant Name: Lorence N'drea

Signature: Lorence

Date: 30th August 2024

Thank you for your valuable contribution!

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Participant Name: Vibha Patil

Signature: Vibha Patil

Date: 30th August 2024

Thank you for your valuable contribution!

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Participant Name: Daniel Chapman

Signature: Daniel

Date: 30th August 2024

Thank you for your valuable contribution!

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Participant Name: Meit Vyas

Signature: Meit Vyas

Date: 30th August 2024

Thank you for your valuable contribution!